```
1  //=========================
2  //simulated Temperature control Process
3
4
5
6  //Calculate target temperature from controller output
7  SIM_TargetTemp:= SIM_AmbientTemp +
8  ((TIC_CV / 100.0) * SIM_HeaterGain * 100.0);
9
10 //First order lag process model
11 TIC_PV:= TIC_PV +
12 ((SIM_TargetTemp - TIC_PV) / SIM_TauSeconds) * SIM_TaskDT;
13
14 //clamp PV to realistic limits
15 if TIC_PV > SIM_MaxTemp
16 then TIC_PV:= SIM_MaxTemp;
17     end_if;
18
19     if TIC_PV < SIM_MinTemp then
20         TIC_PV:= SIM_MinTemp;
21         end_if;
22
23
```